
3 Overview

Fall Armyworm (*Spodoptera frugiperda*) is a destructive pest affecting a wide range of crops in Mozambique. This guide outlines a comprehensive treatment process using Integrated Pest Management (IPM) principles.

3 Affected Crop Families

- **Asteraceae**
- **Capparaceae**
- **Chenopodiaceae**
- **Malvaceae**
- **Rubiaceae**
- **Tiliaceae**
- **Verbenaceae**
- **Commelinaceae**
- **Cyperaceae**

While FAW primarily targets maize and grasses, these families may be affected in mixed cropping systems or under stress.

3 Step-by-Step Treatment Process

1. Prevention and Cultural Control

- **Early Planting:** Plant crops at the beginning of the rainy season.
- **Crop Rotation:** Rotate with non-host crops like legumes.
- **Intercropping:** Use beans or cassava to reduce FAW spread.
- **Field Hygiene:** Remove weeds and crop residues.
- **Resistant Varieties:** Use FAW-resistant or early-maturing varieties.

2. Monitoring and Scouting

- **Start Early:** Begin scouting at seedling stage.
- **Sampling:** Inspect 10 × 10 plants in 5 field locations.
- **Signs:**
 - Window pane leaf damage

- Larval frass in whorls
- Larvae with inverted "Y" on head and four dots on rear

3. Threshold-Based Action

- **Early Whorl Stage:** Treat if >20% of plants are damaged.
- **Late Whorl Stage:** Treat if >40% of plants are freshly damaged.
- **Tassel/Silk Stage:** Avoid spraying.

4. Biological Control

- **Natural Enemies:**
 - Lady beetles
 - *Trichogramma* spp.
 - *Bacillus thuringiensis* (Bt)
- **Habitat Management:** Plant flowering borders to attract beneficial insects.

5. Chemical Control

- **Recommended Insecticides:**
 - Pirimiphos-methyl + Deltamethrin
 - Emamectin benzoate
 - Profenophos + Cypermethrin
- **Application Tips:**
 - Use correct dosage (+5% above recommended rate)
 - Rotate insecticides to prevent resistance
 - Wear protective equipment

6. Integrated Pest Management (IPM)

Combine:

- Agroecological practices
- Biological control agents
- Judicious pesticide use
- Farmer training and awareness

3 Notes for Specific Crop Families

- Monitor adjacent fields and mixed cropping systems.
 - Apply IPM principles consistently.
 - Consult local extension services for crop-specific recommendations.
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3 Resources

- Ministry of Agriculture and Rural Development (Mozambique)
- FAO and CABI IPM Guidelines
- IRRIGA Project Pest Management Plan